

Multiple Choice (10 marks)

1. In general, animal cells differ from plant cells in that animal cells have
 - (a) a cell wall made of cellulose
 - (b) lysosomes
 - (c) large vacuoles that store water
 - (d) centrioles within centrosomes
2. Chemical substances released by organisms that elicit a physiological or behavioral response in other members of the same species are known as
 - (a) auxins
 - (b) neurotransmitters
 - (c) enzymes
 - (d) hormones
 - (e) pheromones
3. Sea divers are aware of a danger known as the "bends". Explain the physiological mechanism of the bends.
 - (a) Carbon dioxide bubbles forming in the blood causing acidosis
 - (b) Rapid increase in barometric pressure causing the bends
 - (c) Formation of helium bubbles in the bloodstream due to rapid ascent
 - (d) Increased salinity in the water leading to dehydration and joint pain
 - (e) Oxygen bubbles forming in the blood leading to hypoxia
4. All of the following play a role in morphogenesis EXCEPT
 - (a) homeotic genes
 - (b) cell division
 - (c) signal transduction
 - (d) cell differentiation
 - (e) apoptosis
5. The polytene chromosomes seen in *Drosophila silvarentis* with the aid of a light microscope are
 - (a) multiple copies of a single chromosome
 - (b) not visible even under a light microscope

- (c) produced only during mitosis
- (d) produced by repeated rounds of nuclear division without DNA replication
- (e) single copies of multiple chromosomes

True / False (10 marks)

Answer True or False

6. True or False: After CLASS—Is a voluntary public insurance program a realistic way to meet the long-term support and service needs of adults with disabilities?
7. True or False: Nutritional assessment of gluten-free diet. Is gluten-free diet deficient in some nutrient?
8. True or False: Is atropine needed with ketamine sedation?
9. True or False: Does frozen section alter surgical management of multinodular thyroid disease?
10. True or False: Learning needs of postpartum women: does socioeconomic status matter?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the meanings of 'gymnosperm' and 'angiosperm,' and analyze the distinguishing characteristics that separate conifers from flowering plants in terms of their adaptations and reproductive strategies.
12. Discuss the functions and significance of the M, G₀, G₁, S, and G₂ phases in the cell cycle, emphasizing their roles in cell growth, organelle duplication, and preparation for division.

End of Paper